

DATA MINING ASSIGNMENT

Kaggle Competition – Data Mining 252 Stage 2

As part of the course assessment, students are required to participate in the Kaggle competition: <https://www.kaggle.com/t/b32672e2d8b44bb696bc50eaea3005f8>

1. Assignment Requirements

Each student/group is required to participate in the competition and submit the following materials:

1.1 Presentation Video

A presentation video (3-5 mins) describing:

- problem understanding,
- data preprocessing methods,
- feature engineering techniques,
- model development and training,
- validation strategy,
- experimental results and discussion.

1.2 Private Leaderboard Ranking Screenshot

A screenshot clearly showing:

- team/student name,
- private leaderboard ranking,
- final private leaderboard score.

1.3 Final Report

A formal report summarizing the entire project, including:

- introduction,
- methodology,
- data preprocessing,
- feature engineering,
- model architecture,
- experimental setup,

- evaluation and analysis,
- conclusion and future work.

All required materials must be submitted through the LMS.

2. Submission Schedule

Phase 1 – Competition Participation

Deadline: 30/05/2026

Students are required to:

- participate in the Kaggle competition,
- submit prediction results to the competition leaderboard,
- experiment with submissions on both the public and private test samples.

Phase 2 – Final Submission Selection

Deadline: 03/06/2026

Students must finalize and select **three (3) submissions** for evaluation on the private leaderboard.

The final competition ranking used for grading will be determined based on the private leaderboard results.

Phase 3 – Final Submission on LMS

Deadline: 07/06/2026

Submit every required files to the LMS

3. Assessment Criteria

The final assignment score will be computed as follows:

$$\text{Final Score} = 0.6 \times \text{Ranking Score} + 0.4 \times \text{Report Score}$$

3.1 Ranking Score (60%)

The ranking score is determined according to the team's/private leaderboard performance.

Scores will be normalized using a normal distribution with:

- Maximum score: **10**
- Minimum score: **7**

Higher leaderboard rankings will receive proportionally higher scores.

3.2 Report Score (40%)

The report will be evaluated based on:

- technical correctness,
- quality of methodology,
- experimental design,
- analytical discussion,
- clarity of presentation,
- overall completeness.

4. Important Notes

- Students are encouraged to explore different machine learning and data mining approaches, including ensemble methods, feature engineering, and model optimization techniques.
- All submissions must represent the students' own work.
- Any form of plagiarism or unauthorized sharing of solutions is strictly prohibited.
- Late submissions may be subject to penalties in accordance with course regulations.
- The private leaderboard score will be considered the official competition result for grading purposes.